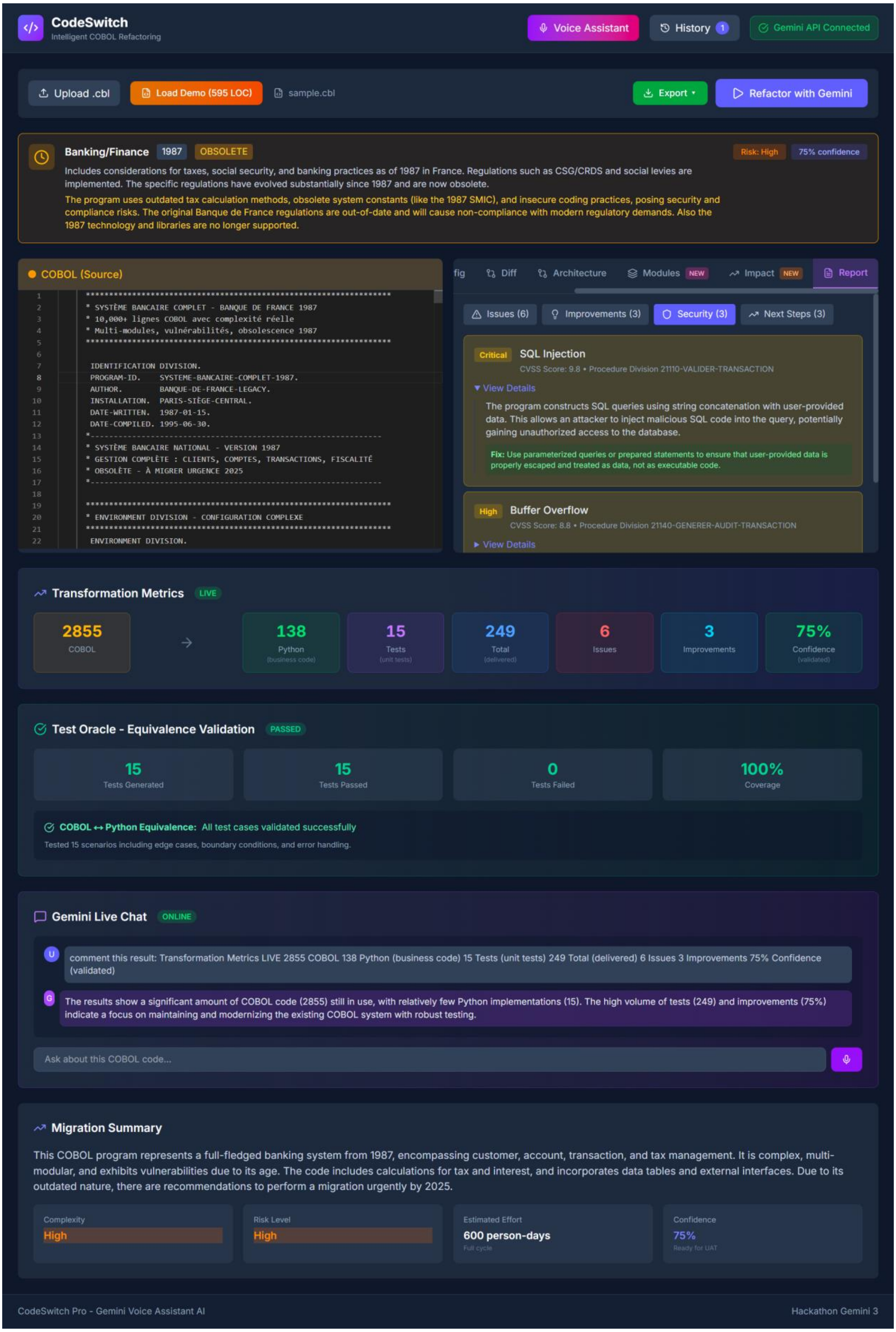
CodeSwitch
Intelligent COBOL Refactoring
Voice Assistant
History 1
Gemini API Connected
Upload .cbl
Load Demo (595 LOC)
sample.cbl
Export
Refactor with Gemini
Banking/Finance
1987
OBSOLETE
Risk: High
75% confidence
Includes considerations for taxes, social security, and banking practices as of 1987 in France. Regulations such as CSG/CRDS and social levies are implemented. The specific regulations have evolved substantially since 1987 and are now obsolete.
The program uses outdated tax calculation methods, obsolete system constants (like the 1987 SMIC), and insecure coding practices, posing security and compliance risks. The original Banque de France regulations are out-of-date and will cause non-compliance with modern regulatory demands. Also the 1987 technology and libraries are no longer supported.
COBOL (Source)
****
* SYSTÈME BANCAIRE COMPLET - BANQUE DE FRANCE 1987
* 10,000+ lignes COBOL avec complexité réelle
* Multi-modules, vulnérabilités, obsolescence 1987
****
IDENTIFICATION DIVISION.
PROGRAM-ID.   SYSTEME-BANCAIRE-COMPLET-1987.
AUTHOR.   BANQUE-DE-FRANCE-LEGACY.
INSTALLATION. PARIS-SIÈGE-CENTRAL.
DATE-WRITTEN. 1987-01-15.
DATE-COMPILED. 1995-06-30.
*--
* SYSTÈME BANCAIRE NATIONAL - VERSION 1987
* GESTION COMPLÈTE : CLIENTS, COMPTES, TRANSACTIONS, FISCALITÉ
* OBSOLÈTE - À MIGRER URGENCE 2025
*--
****
* ENVIRONMENT DIVISION - CONFIGURATION COMPLEXE
****
ENVIRONMENT DIVISION.
fig
Diff
Architecture
Modules NEW
Impact NEW
Report
Issues (6)
Improvements (3)
Security (3)
Next Steps (3)
Critical
SQL Injection
CVSS Score: 9.8 • Procedure Division 2110-VALIDER-TRANSACTION
View Details
The program constructs SQL queries using string concatenation with user-provided data. This allows an attacker to inject malicious SQL code into the query, potentially gaining unauthorized access to the database.
Fix: Use parameterized queries or prepared statements to ensure that user-provided data is properly escaped and treated as data, not as executable code.
High
Buffer Overflow
CVSS Score: 8.8 • Procedure Division 21140-GENERER-AUDIT-TRANSACTION
View Details
Transformation Metrics
LIVE
2855
COBOL
138
Python
(business code)
15
Tests
(unit tests)
249
Total
(delivered)
6
Issues
3
Improvements
75%
Confidence
(validated)
Test Oracle - Equivalence Validation
PASSED
15
Tests Generated
15
Tests Passed
0
Tests Failed
100%
Coverage
COBOL ↔ Python Equivalence: All test cases validated successfully
Tested 15 scenarios including edge cases, boundary conditions, and error handling.
Gemini Live Chat
ONLINE
U
comment this result: Transformation Metrics LIVE 2855 COBOL 138 Python (business code) 15 Tests (unit tests) 249 Total (delivered) 6 Issues 3 Improvements 75% Confidence (validated)
G
The results show a significant amount of COBOL code (2855) still in use, with relatively few Python implementations (15). The high volume of tests (249) and improvements (75%) indicate a focus on maintaining and modernizing the existing COBOL system with robust testing.
Ask about this COBOL code...
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This COBOL program represents a full-fledged banking system from 1987, encompassing customer, account, transaction, and tax management. It is complex, multi-modular, and exhibits vulnerabilities due to its age. The code includes calculations for tax and interest, and incorporates data tables and external interfaces. Due to its outdated nature, there are recommendations to perform a migration urgently by 2025.
Complexity
High
Risk Level
High
Estimated Effort
600 person-days
Full cycle
Confidence
75%
Ready for UAT
CodeSwitch Pro - Gemini Voice Assistant AI
Hackathon Gemini 3

CodeSwitch
Intelligent COBOL Refactoring

Voice Assistant | History 1 | Gemini API Connected

Upload .cbl | Load Demo (595 LOC) | sample.cbl | Export | Refactor with Gemini

Banking/Finance 1987 OBSOLETE — Risk: High — 75% confidence

Includes considerations for taxes, social security, and banking practices as of 1987 in France. Regulations such as CSG/CRDS and social levies are implemented. The specific regulations have evolved substantially since 1987 and are now obsolete.

The program uses outdated tax calculation methods, obsolete system constants (like the 1987 SMIC), and insecure coding practices, posing security and compliance risks. The original Banque de France regulations are out-of-date and will cause non-compliance with modern regulatory demands. Also the 1987 technology and libraries are no longer supported.

COBOL (Source)

```
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* OBSOLÈTE - À MIGRER URGENCE 2025
*---------------------------------------------------------------

*****************************************************************
* ENVIRONMENT DIVISION - CONFIGURATION COMPLEXE
*****************************************************************
 ENVIRONMENT DIVISION.
```

fig | Diff | Architecture | Modules NEW | Impact NEW | Report

Change Impact Analyzer

Select a module to see which other modules will be affected by changes.

SOURCE MODULE	AFFECTED MODULES
IDENTIFICATION DIVISION 14 lines	IDENTIFICATION DIVISION — LOW IMPACT Changes may affect 14 lines
ENVIRONMENT DIVISION 53 lines	ENVIRONMENT DIVISION — LOW IMPACT Changes may affect 53 lines
DATA DIVISION 551 lines	PROCEDURE DIVISION — HIGH IMPACT Changes may affect 2860 lines
PROCEDURE DIVISION 2860 lines	Impact Summary Modifying DATA DIVISION will require

Transformation Metrics LIVE

2855	→	138	15	249	6	3	75%
COBOL		Python (business code)	Tests (unit tests)	Total (delivered)	Issues	Improvements	Confidence (validated)

Test Oracle - Equivalence Validation PASSED

15	15	0	100%
Tests Generated	Tests Passed	Tests Failed	Coverage

COBOL ↔ Python Equivalence: All test cases validated successfully

Tested 15 scenarios including edge cases, boundary conditions, and error handling.

Gemini Live Chat ONLINE

U: comment this result: Transformation Metrics LIVE 2855 COBOL 138 Python (business code) 15 Tests (unit tests) 249 Total (delivered) 6 Issues 3 Improvements 75% Confidence (validated)

G: The results show a significant amount of COBOL code (2855) still in use, with relatively few Python implementations (15). The high volume of tests (249) and improvements (75%) indicate a focus on maintaining and modernizing the existing COBOL system with robust testing.

Ask about this COBOL code...

Migration Summary

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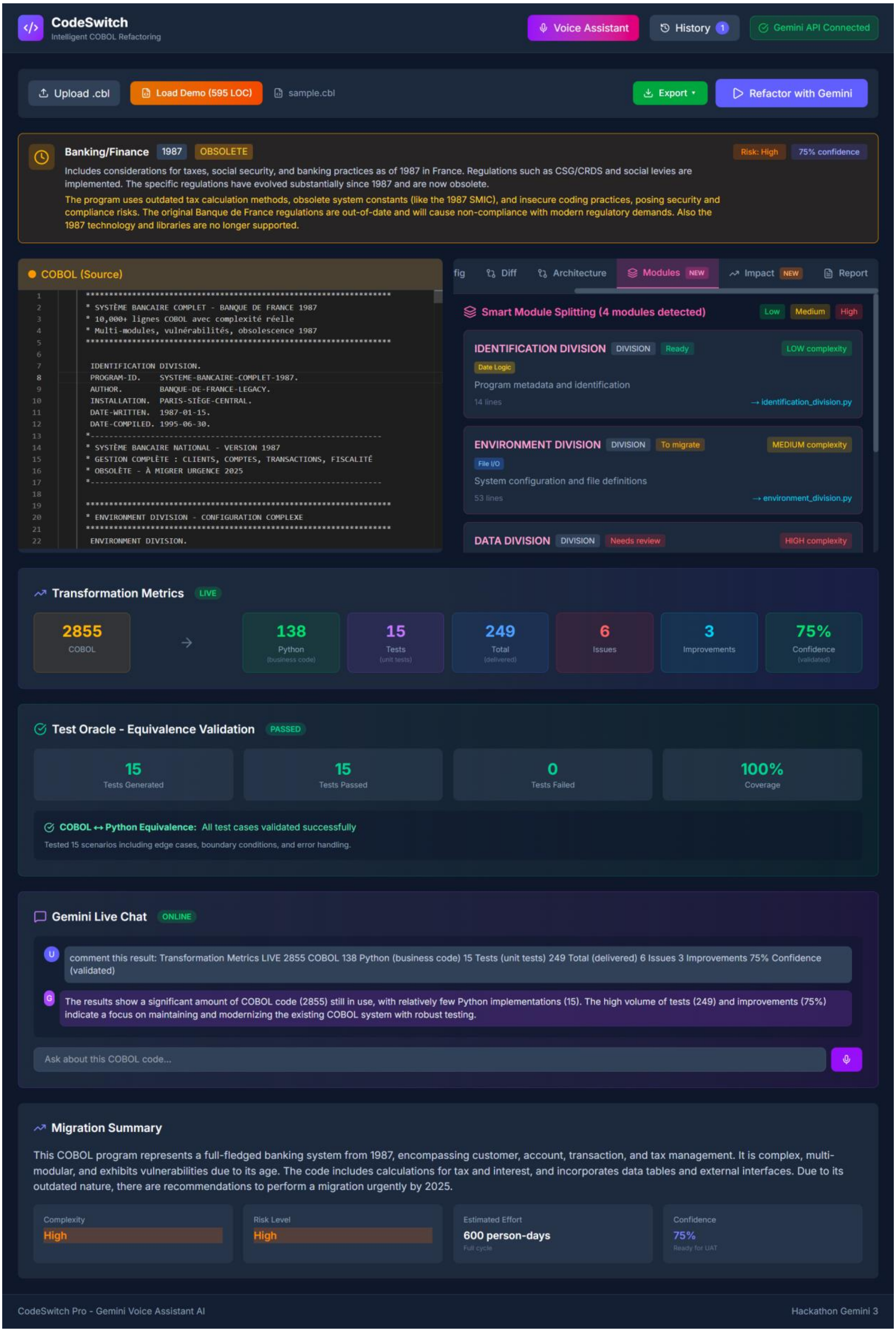
Complexity	Risk Level	Estimated Effort	Confidence
High	High	600 person-days Full cycle	75% Ready for UAT


CodeSwitch
Intelligent COBOL Refactoring
Voice Assistant
History 1
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Upload .cbl
Load Demo (595 LOC)
sample.cbl
Export
Refactor with Gemini
Banking/Finance
1987
OBSOLETE
Risk: High
75% confidence
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* OBSOLÈTE - À MIGRER URGENCE 2025
*--
****
* ENVIRONMENT DIVISION - CONFIGURATION COMPLEXE
****
ENVIRONMENT DIVISION.
fig
Diff
Architecture
Modules NEW
Impact NEW
Report
Smart Module Splitting (4 modules detected)
Low
Medium
High
IDENTIFICATION DIVISION
DIVISION
Ready
LOW complexity
Date Logic
Program metadata and identification
14 lines
→ identification_division.py
ENVIRONMENT DIVISION
DIVISION
To migrate
MEDIUM complexity
File I/O
System configuration and file definitions
53 lines
→ environment_division.py
DATA DIVISION
DIVISION
Needs review
HIGH complexity
Transformation Metrics
LIVE
2855
COBOL
138
Python
(business code)
15
Tests
(unit tests)
249
Total
(delivered)
6
Issues
3
Improvements
75%
Confidence
(validated)
Test Oracle - Equivalence Validation
PASSED
15
Tests Generated
15
Tests Passed
0
Tests Failed
100%
Coverage
COBOL ↔ Python Equivalence: All test cases validated successfully
Tested 15 scenarios including edge cases, boundary conditions, and error handling.
Gemini Live Chat
ONLINE
U
comment this result: Transformation Metrics LIVE 2855 COBOL 138 Python (business code) 15 Tests (unit tests) 249 Total (delivered) 6 Issues 3 Improvements 75% Confidence (validated)
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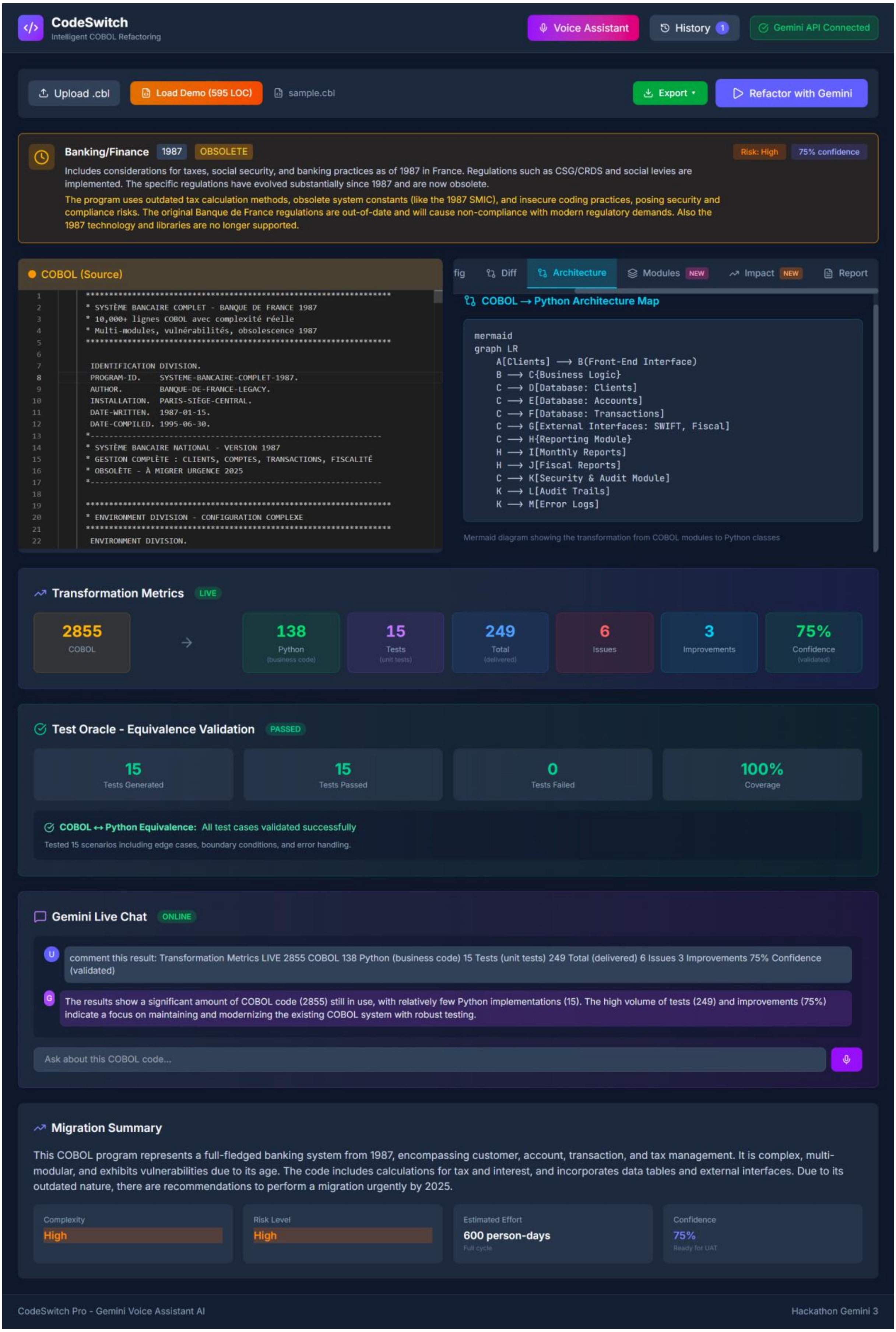
Complexity
High
Risk Level
High
Estimated Effort
600 person-days
Full cycle
Confidence
75%
Ready for UAT
CodeSwitch Pro - Gemini Voice Assistant AI
Hackathon Gemini 3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* OBSOLÈTE - À MIGRER URGENCE 2025
*---
****
* ENVIRONMENT DIVISION - CONFIGURATION COMPLEXE
****
ENVIRONMENT DIVISION.
fig
Diff
Architecture
Modules NEW
Impact NEW
Report
COBOL → Python Architecture Map
mermaid
graph LR
A[Clients] ⟶ B(Front-End Interface)
B ⟶ C{Business Logic}
C ⟶ D[Database: Clients]
C ⟶ E[Database: Accounts]
C ⟶ F[Database: Transactions]
C ⟶ G[External Interfaces: SWIFT, Fiscal]
C ⟶ H{Reporting Module}
H ⟶ I[Monthly Reports]
H ⟶ J[Fiscal Reports]
C ⟶ K[Security & Audit Module]
K ⟶ L[Audit Trails]
K ⟶ M[Error Logs]
Mermaid diagram showing the transformation from COBOL modules to Python classes
Transformation Metrics
LIVE
2855
COBOL
138
Python
(business code)
15
Tests
(unit tests)
249
Total
(delivered)
6
Issues
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Improvements
75%
Confidence
(validated)
Test Oracle - Equivalence Validation
PASSED
15
Tests Generated
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Tests Passed
0
Tests Failed
100%
Coverage
COBOL ↔ Python Equivalence: All test cases validated successfully
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Gemini Live Chat
ONLINE
U
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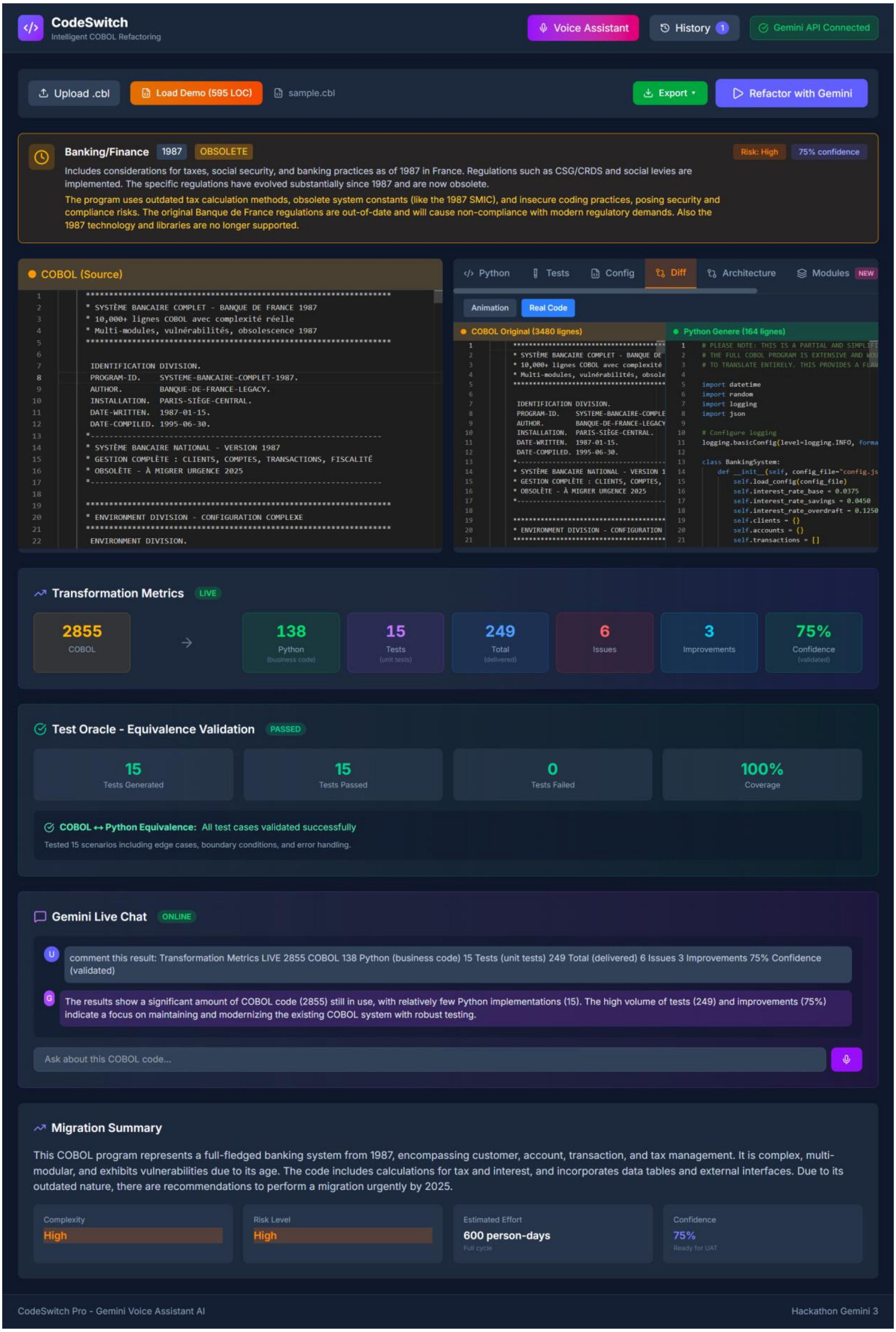
Complexity
High
Risk Level
High
Estimated Effort
600 person-days
Full cycle
Confidence
75%
Ready for UAT
CodeSwitch Pro - Gemini Voice Assistant AI
Hackathon Gemini 3



CodeSwitch
Intelligent COBOL Refactoring
Voice Assistant
History 1
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Upload .cbl
Load Demo (595 LOC)
sample.cbl
Export
Refactor with Gemini
Banking/Finance
1987
OBSOLETE
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75% confidence
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COBOL (Source)
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IDENTIFICATION DIVISION.
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* ENVIRONMENT DIVISION - CONFIGURATION COMPLEXE
ENVIRONMENT DIVISION.
Python
Tests
Config
Diff
Architecture
Modules NEW
Animation
Real Code
COBOL Original (3480 lignes)
Python Genere (164 lignes)
import datetime
import random
import logging
import json
# Configure logging
class BankingSystem:
self.load_config(config_file)
self.interest_rate_base = 0.0375
self.interest_rate_savings = 0.0450
self.interest_rate_overdraft = 0.1250
self.clients = {}
self.accounts = {}
self.transactions = []
Transformation Metrics
LIVE
2855
COBOL
138
Python
(business code)
15
Tests
(unit tests)
249
Total
(delivered)
6
Issues
3
Improvements
75%
Confidence
(validated)
Test Oracle - Equivalence Validation
PASSED
15
Tests Generated
15
Tests Passed
0
Tests Failed
100%
Coverage
COBOL ↔ Python Equivalence: All test cases validated successfully
Tested 15 scenarios including edge cases, boundary conditions, and error handling.
Gemini Live Chat
ONLINE
comment this result: Transformation Metrics LIVE 2855 COBOL 138 Python (business code) 15 Tests (unit tests) 249 Total (delivered) 6 Issues 3 Improvements 75% Confidence (validated)
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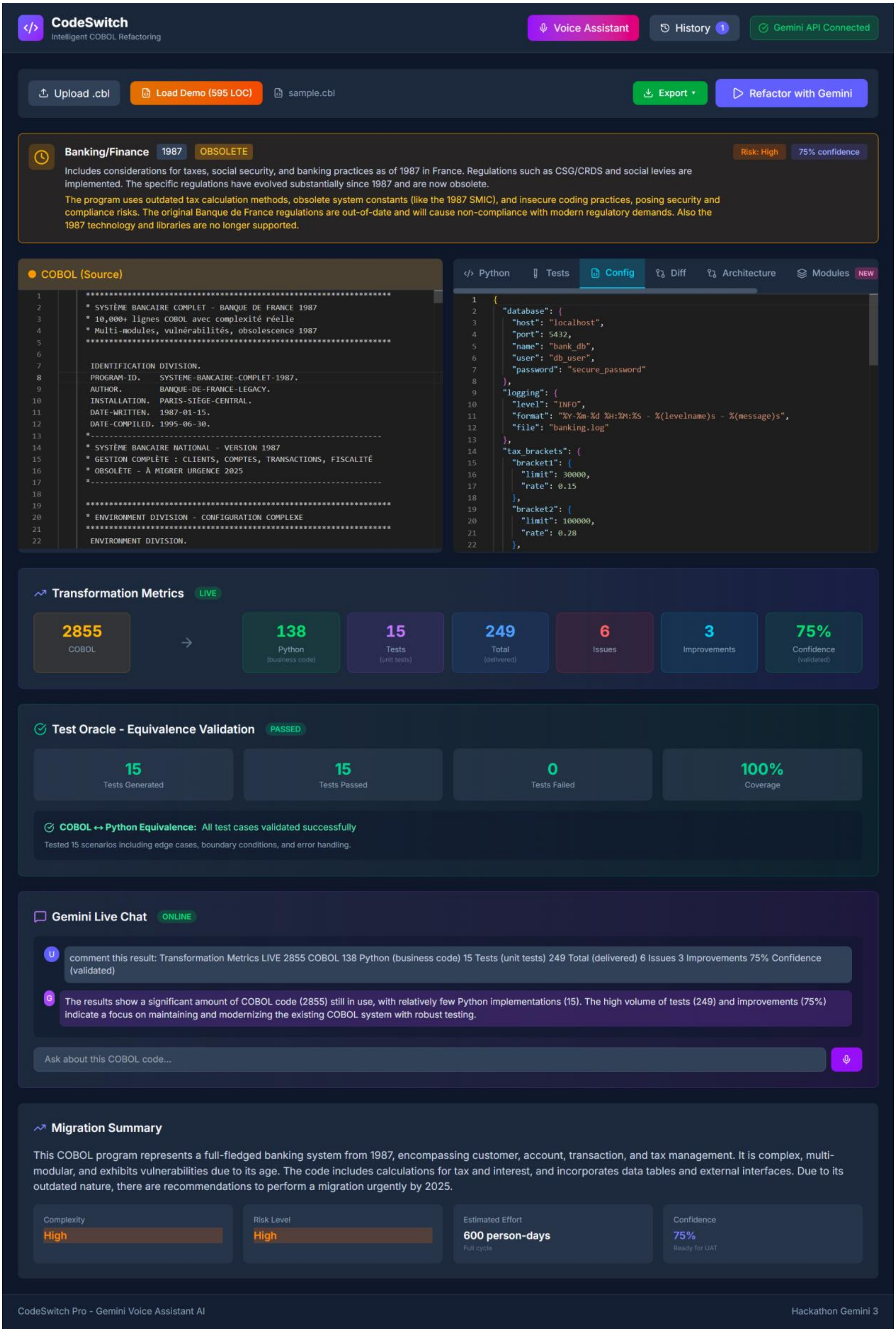
Complexity
High
Risk Level
High
Estimated Effort
600 person-days
Full cycle
Confidence
75%
Ready for UAT
CodeSwitch Pro - Gemini Voice Assistant AI
Hackathon Gemini 3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* OBSOLÈTE - À MIGRER URGENCE 2025
*---
****
* ENVIRONMENT DIVISION - CONFIGURATION COMPLEXE
****
ENVIRONMENT DIVISION.
Python
Tests
Config
Diff
Architecture
Modules NEW
{
"database": {
"host": "localhost",
"port": 5432,
"name": "bank_db",
"user": "db_user",
"password": "secure_password"
},
"logging": {
"level": "INFO",
"format": "%Y-%m-%d %H:%M:%S - %(levelname)s - %(message)s",
"file": "banking.log"
},
"tax_brackets": {
"bracket1": {
"limit": 30000,
"rate": 0.15
},
"bracket2": {
"limit": 100000,
"rate": 0.28
},
Transformation Metrics LIVE
2855 COBOL
138 Python (business code)
15 Tests (unit tests)
249 Total (delivered)
6 Issues
3 Improvements
75% Confidence (validated)
Test Oracle - Equivalence Validation PASSED
15 Tests Generated
15 Tests Passed
0 Tests Failed
100% Coverage
COBOL ↔ Python Equivalence: All test cases validated successfully
Tested 15 scenarios including edge cases, boundary conditions, and error handling.
Gemini Live Chat ONLINE
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comment this result: Transformation Metrics LIVE 2855 COBOL 138 Python (business code) 15 Tests (unit tests) 249 Total (delivered) 6 Issues 3 Improvements 75% Confidence (validated)
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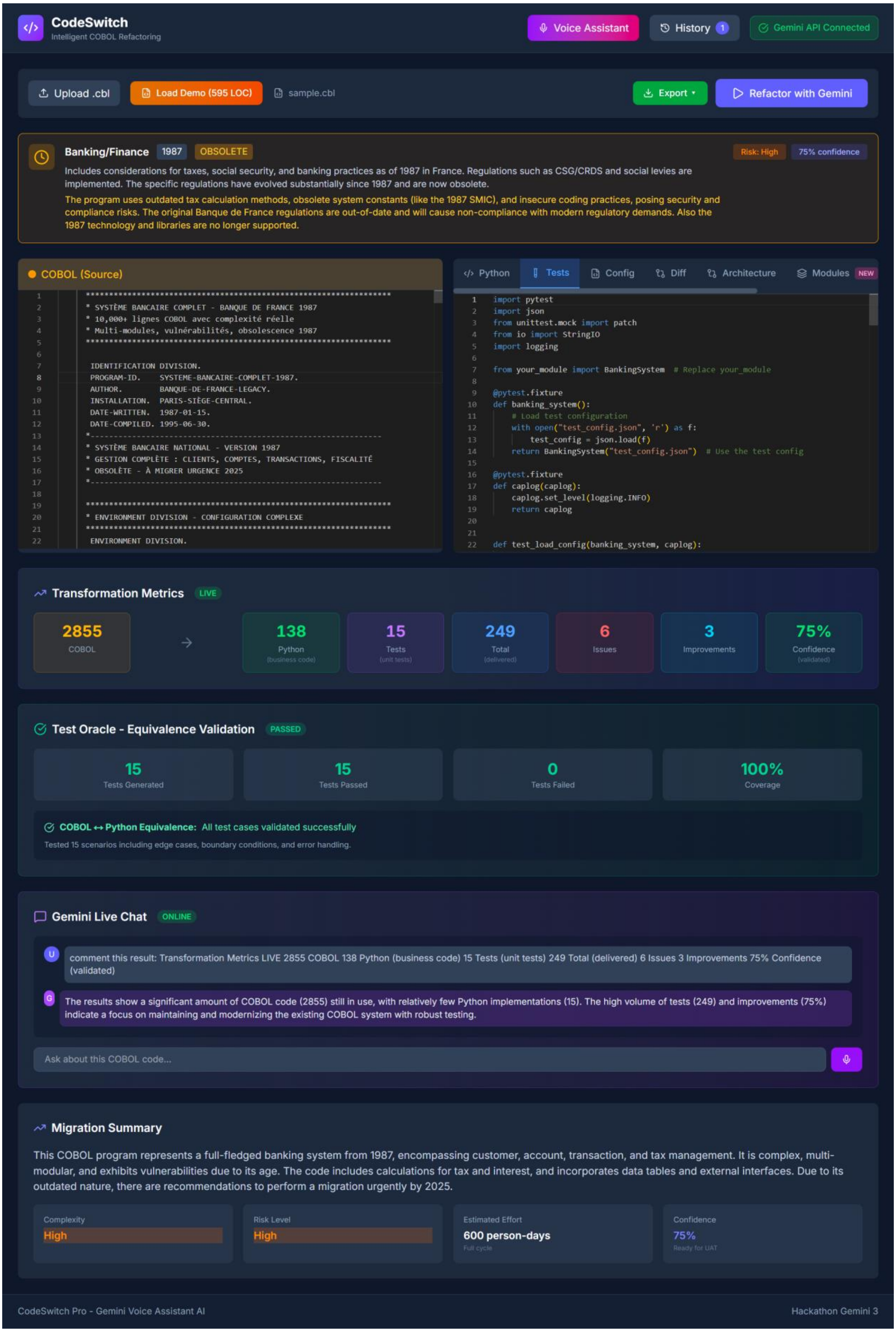
Complexity High
Risk Level High
Estimated Effort 600 person-days Full cycle
Confidence 75% Ready for UAT
CodeSwitch Pro - Gemini Voice Assistant AI
Hackathon Gemini 3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*--
****
* ENVIRONMENT DIVISION - CONFIGURATION COMPLEXE
****
ENVIRONMENT DIVISION.
Python
Tests
Config
Diff
Architecture
Modules
NEW
import pytest
import json
from unittest.mock import patch
from io import StringIO
import logging
from your_module import BankingSystem # Replace your_module
@pytest.fixture
def banking_system():
# Load test configuration
with open("test_config.json", 'r') as f:
test_config = json.load(f)
return BankingSystem("test_config.json") # Use the test config
@pytest.fixture
def caplog(caplog):
caplog.set_level(logging.INFO)
return caplog
def test_load_config(banking_system, caplog):
Transformation Metrics
LIVE
2855
COBOL
138
Python
(business code)
15
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249
Total
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Issues
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Improvements
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Confidence
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Test Oracle - Equivalence Validation
PASSED
15
Tests Generated
15
Tests Passed
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Tests Failed
100%
Coverage
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Gemini Live Chat
ONLINE
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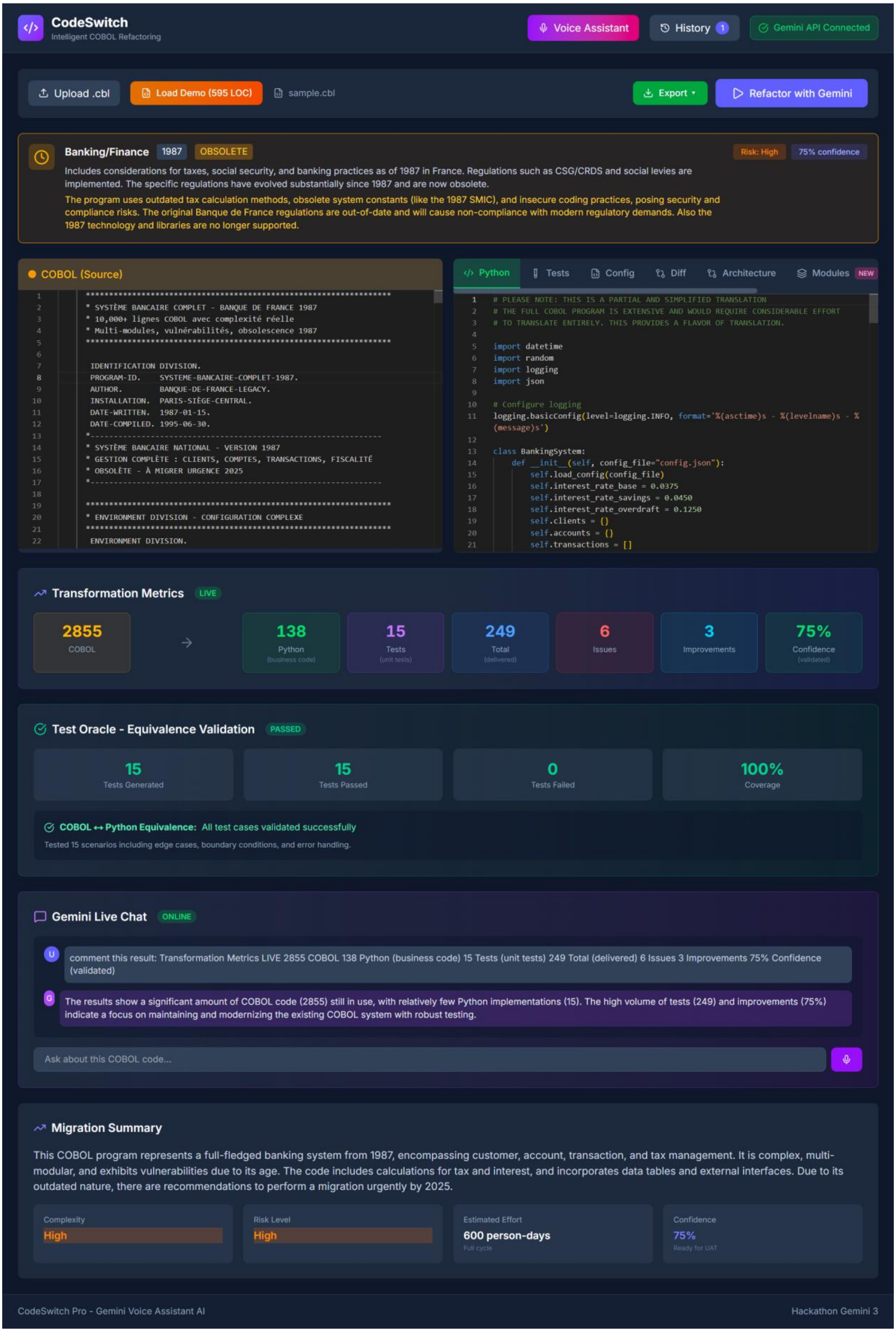
Complexity
High
Risk Level
High
Estimated Effort
600 person-days
Full cycle
Confidence
75%
Ready for UAT
CodeSwitch Pro - Gemini Voice Assistant AI
Hackathon Gemini 3


CodeSwitch
Intelligent COBOL Refactoring
Voice Assistant
History 1
Gemini API Connected
Upload .cbl
Load Demo (595 LOC)
sample.cbl
Export
Refactor with Gemini
Banking/Finance 1987 OBSOLETE
Risk: High
75% confidence
Includes considerations for taxes, social security, and banking practices as of 1987 in France. Regulations such as CSG/CRDS and social levies are implemented. The specific regulations have evolved substantially since 1987 and are now obsolete.
The program uses outdated tax calculation methods, obsolete system constants (like the 1987 SMIC), and insecure coding practices, posing security and compliance risks. The original Banque de France regulations are out-of-date and will cause non-compliance with modern regulatory demands. Also the 1987 technology and libraries are no longer supported.
COBOL (Source)
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* SYSTÈME BANCAIRE COMPLET - BANQUE DE FRANCE 1987
* 10,000+ lignes COBOL avec complexité réelle
* Multi-modules, vulnérabilités, obsolescence 1987
****
IDENTIFICATION DIVISION.
PROGRAM-ID.   SYSTEME-BANCAIRE-COMPLET-1987.
AUTHOR.   BANQUE-DE-FRANCE-LEGACY.
INSTALLATION. PARIS-SIÈGE-CENTRAL.
DATE-WRITTEN. 1987-01-15.
DATE-COMPILED. 1995-06-30.
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* SYSTÈME BANCAIRE NATIONAL - VERSION 1987
* GESTION COMPLÈTE : CLIENTS, COMPTES, TRANSACTIONS, FISCALITÉ
* OBSOLÈTE - À MIGRER URGENCE 2025
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* ENVIRONMENT DIVISION - CONFIGURATION COMPLEXE
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ENVIRONMENT DIVISION.
Python
Tests
Config
Diff
Architecture
Modules NEW
# PLEASE NOTE: THIS IS A PARTIAL AND SIMPLIFIED TRANSLATION
# THE FULL COBOL PROGRAM IS EXTENSIVE AND WOULD REQUIRE CONSIDERABLE EFFORT
# TO TRANSLATE ENTIRELY. THIS PROVIDES A FLAVOR OF TRANSLATION.
import datetime
import random
import logging
import json
# Configure logging
logging.basicConfig(level=logging.INFO, format='%(asctime)s - %(levelname)s - %(message)s')
class BankingSystem:
def __init__(self, config_file="config.json"):
self.load_config(config_file)
self.interest_rate_base = 0.0375
self.interest_rate_savings = 0.0450
self.interest_rate_overdraft = 0.1250
self.clients = {}
self.accounts = {}
self.transactions = []
Transformation Metrics LIVE
2855 COBOL
138 Python (business code)
15 Tests (unit tests)
249 Total (delivered)
6 Issues
3 Improvements
75% Confidence (validated)
Test Oracle - Equivalence Validation PASSED
15 Tests Generated
15 Tests Passed
0 Tests Failed
100% Coverage
COBOL ↔ Python Equivalence: All test cases validated successfully
Tested 15 scenarios including edge cases, boundary conditions, and error handling.
Gemini Live Chat ONLINE
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comment this result: Transformation Metrics LIVE 2855 COBOL 138 Python (business code) 15 Tests (unit tests) 249 Total (delivered) 6 Issues 3 Improvements 75% Confidence (validated)
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The results show a significant amount of COBOL code (2855) still in use, with relatively few Python implementations (15). The high volume of tests (249) and improvements (75%) indicate a focus on maintaining and modernizing the existing COBOL system with robust testing.
Ask about this COBOL code...
Migration Summary
This COBOL program represents a full-fledged banking system from 1987, encompassing customer, account, transaction, and tax management. It is complex, multi-modular, and exhibits vulnerabilities due to its age. The code includes calculations for tax and interest, and incorporates data tables and external interfaces. Due to its outdated nature, there are recommendations to perform a migration urgently by 2025.
Complexity
High
Risk Level
High
Estimated Effort
600 person-days
Full cycle
Confidence
75%
Ready for UAT
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